The Nested Partner Ring Read as a Folded Square: A Rolling Stack, Four Providers, a Straight-Line Gradient and a Square-Root Address

Driven by Dean A. Kulik
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Abstract

The nested partner ring is Dean Kulik's construction. Reverse a binary word, unless it is a palindrome, then complement it. Place the smaller member of every pair in ascending order on one half of a circle, set each partner directly opposite its word, and swap the two words that meet at the centre seam. The ring is never held. It is addressed.

This paper holds the finished state first and reads everything back from it. Cut each word into its front half and its back half and plot it at the point whose coordinates are the front half and the reversed back half. Representatives fill the triangle above the diagonal of the unit square, partners fill its mirror image below, and palindromes sit on the diagonal, which is itself folded at its midpoint. The square is folded twice, and that double fold is the whole ring.

Ten routes run back from the fold, and every one is an algorithm. The spreadsheet's rolling columns are a two-provider stack, so the counter-clockwise column is the clockwise column three rounds later and the 8-bit cross pairs are the half turn. The ring is half built and half reflected, with a twist confined to its palindromic words. Every ring is the 2-bit ring used as a frame alphabet around the ring two lengths below: four providers, one of

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which is continuation, so continuation depth falls as 4^{-t} and averages $1/3$. The representative density is the straight line $1 - t$, so a word of value t sits at angle $t - t^2/2$ plus a tent that is the fold of the diagonal. The seam is the apex where the density reaches zero. The complement branch compiles toward value $1/3$ at angle $5/18$. Rank and unrank need one quadratic, one integer square root and one bit-reversal count. The radial reader reads the digits of $1 - \sqrt{1 - 2u}$ in the front half of every word. Merging each word with its partner leaves grey exactly where the two disagree: all of it at two bits, half of it in the limit, and all of it again on the diagonal and the anti-diagonal.

Twenty-eight lines were sealed before their checks ran. Twenty-seven run as sealed. The twenty-eighth runs at every width except the degenerate width 1, where it lands elsewhere and is routed to its owner. Two claims from the previous session are withdrawn and routed. The open boundary is named in the ledger, Section 14.

1. The held state

A word x of width L is an integer written most significant bit first. Write J for reversal and C for complement. The partner is

$$P(x) = \begin{cases} J(x), & J(x) \neq x, \\ C(x), & J(x) = x. \end{cases}$$

P is an involution with no fixed point: a palindrome is never its own complement and a non-palindrome is never its own reversal. This is the C1 axiom in its smallest form, since no word may hold its place without a partner to move to. The 2^L words fall into 2^{L-1} pairs. The representative of a pair is its smaller member. Representatives run in ascending order on positions 0 to $M - 1$, with $N = 2^L$ and $M = N/2$. The partner of the representative at position i sits at $N - 1 - i$, and positions $M - 1$ and M are swapped. The tape T_L is the ring read as $L \cdot 2^L$ slices. The spreadsheet's "sheet" reading, the circle read backwards with 0 and 1 swapped, is $S_L = C\big(J(T_L)\big)$. At two bits the circle is $T_2 = 00\,10\,01\,11$ and the sheet is $S_2 = 00011011$.

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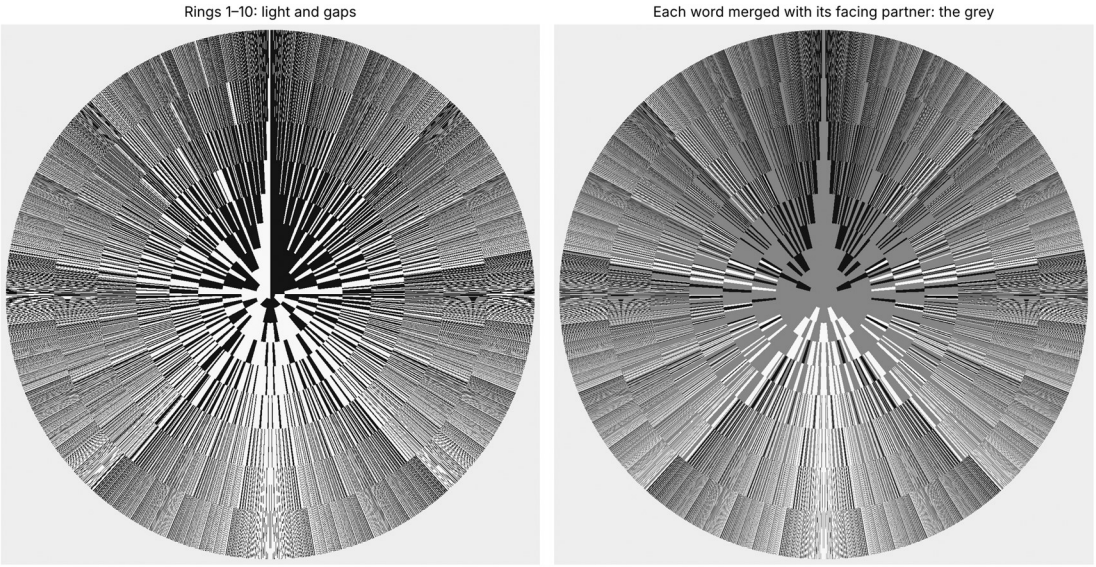


Figure 1. Rings 1 to 10, with radial thickness equal to word length and angle measured clockwise from the top. Left: light (1) and gaps (0). Right: every word merged with its facing partner, slice by slice, which leaves grey wherever the pair disagrees.

The finished state is held before any route is walked. For even $L = 2n$, split $x = f \parallel g$ into a front half f and a back half g of n bits each, and place x at

$$(\varphi, \gamma) = \left(\frac{f}{2^n}, \frac{J(g)}{2^n} \right).$$

The representatives are the cells strictly above the diagonal ($\gamma > \varphi$) together with the lower half of the diagonal. The partner of an off-diagonal cell is its mirror cell (γ, φ). The partner of a diagonal cell, a palindrome, is the diagonal antipode $1 - \varphi$. The square is folded along its diagonal, and the diagonal is folded at its midpoint. Every law below is a reading of that double fold. For odd $L = 2n + 1$ the middle bit is a free coordinate that doubles every cell, so the square carries two sheets.

The routes were walked from inside this space, and each trace is the algorithm that walked it. Every prediction was sealed with SHA-256 before its check ran. The executed notebook nested_ring_routes.ipynb keeps every trace, and its ledger prints one line per sealed claim. Ledger words follow the programme: STANDS (supplied by a theorem or a measured fact), RUNS (the route computed the landing before the data was linked), DEAD END (ran, landed elsewhere, routed to its owner), OPEN (sealed, a chain link not yet written) and PULL (an interpretive bridge).

2. Route 1. The sheet is a stack

The held state. The window table in two_bit_circle.xlsx is a two-dimensional shift register. Rows are slice offsets, rounds are window widths, and the two axes are one axis read diagonally.

The trace. Rebuild every cell from the sheet's own rules. A window of width W is read clockwise from row r. The "counter-clockwise" column is the same clockwise window started three slices later, at the slice where

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the first 11 lands. Hex nib joins two rows. All 896 cached cells of the seven blocks rebuild exactly (RUNS). Then read one row down and one round up at the same time.

What fell out. A window $w_i(W)$ on any cyclic tape s has two providers, and they always agree:

$$w_i(W+1) = 2^W s_i + w_{i+1}(W) = 2\,w_i(W) + s_{i+W}.$$

The first provider pushes the slice being left behind onto the top of the window one row down. The second appends the entering slice at the bottom. Each column therefore slides up one row per round, carrying its stack and pushing one more bit on top. This is the DI chain the spreadsheet shows in columns B and G. Column B by round, rows 0 to 7, reads $(0,0,1,3,2,1,3,2)$, then $(0,1,3,6,5,3,6,4)$, $(1,3,6,13,11,6,12,8)$, $(3,6,13,27,22,12,24,17)$, $(6,13,27,54,44,24,49,35)$, $(13,27,54,108,88,49,99,70)$ and $(27,54,108,216,177,99,198,141)$. Row 0 follows $c_{W+1} = 2c_W + s_W$ and reads $0,0,1,3,6,13,27,54,108,216$.

Column G is column B three rows on. The sheet opens with three dark slices, $s_0 s_1 s_2 = 000$, so three rows on is three rounds later:

$$G_0(W) = B_0(W+3).$$

The counter-clockwise chain $3,6,13,27,54,108,216$ that starts at two bits drops into the clockwise column at five bits and runs in both columns from there (RUNS).

The nibbles carry the same law. The low nibble of any window is the 4-bit window $W-4$ rows down. Before five bits every value is one nibble. At five bits a second nibble exists for the first time, and it holds the single pushed bit. At eight bits both nibbles are full cells of the 4-bit field, half a turn apart:

$$w_i(8) = 16\,w_i(4) + w_{i+4}(4).$$

The half turn of an 8-bit window is therefore the nibble swap. That is the content of the sheet's marked cross pairs: each Hex nib cell meets its own nibble swap one row down and half a turn around, giving $(1B36,\ B163)$, $(6CD8,\ C68D)$, $(B163,\ 1B36)$ and $(C68D,\ 6CD8)$ (RUNS). The purple pairs at six and seven bits are the three-row offset itself.

The 2-bit sheet also reflects at every width, not only at eight. Reversing 00011011 gives the same string rotated by three, so $\mathrm{Rev}_W\, w_i(W) = w_{(11-i-W) \bmod 8}(W)$ for every W from 1 to 8 (RUNS).

Beyond two bits. The stack laws hold verbatim on every tape and every sheet built for $L = 2$ to 5 (RUNS). The half turn at the circumference swaps the two halves of the window on any tape; on ring 3 it swaps the 12-bit halves of the 24-slice window (RUNS). The pure reflection does not carry past two bits. For $L \geq 3$ the tape begins with the unique run 0^{2L-1}, followed by a single 1 in T_L and by 1^L in its reverse, so the reverse of T_L is never a rotation of T_L (RUNS for $L = 3$ to 10). The reflection survives with a twist, and the twist is Route 2. The companion workbook carries the same sheet for ring 3 at every width from 2 to 24 and for ring 4 from 2 to 16.

A frame is one push and one append, $w_i(W+2) = 2^{W+1} s_i + 2\,w_{i+1}(W) + s_{i+W+1}$. A window wrapped in its two outer slices is the rolling stack run twice. Route 3 runs the same path across rings instead of across windows.

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3. Route 2. Half built, half reflected

The held state. Only positions 0 to $M-1$ are built; every other position is a reflection.

The trace. Write each position as the partner of the position facing it, then read the tape slice by slice.

What fell out. At word level $R_L[N-1-i] = P(R_L[i])$, checked at all 32,766 positions for $L \le 14$. At slice level the tape is a palindrome everywhere except on palindromic words, where it is anti-palindromic:

$$T_L[LN-1-p] = T_L[p] \oplus \text{pal(word holding } p),$$

checked at all 196,610 slices for $L \le 13$ (RUNS). The twist covers $2^{\lceil L/2 \rceil}$ of the 2^L words, a share of $1/2$ at widths 2 and 3, $1/4$ at 4 and 5, and halving every two widths after that. At two bits the twist hides inside a rotation, which is why the 2-bit sheet reflects purely; past two bits it shows.

The sheet reading is the tape with every non-palindromic word complemented (RUNS). At two bits that turns $00\ 10\ 01\ 11$ into $00\ 01\ 10\ 11$, which is counting order. That coincidence is the owner of the previous session's identification of the rings with counting: the sheet of the 2-bit ring is the counter, and for every $L \ge 3$ the sheet is not (RUNS, $L = 3$ to 12). Section 13 routes the claim.

4. Route 3. Four providers

The held state. Every ring is the 2-bit ring used as a frame alphabet around the ring two lengths below.

The trace. Peel the outer pair of bits and ask what the middle must satisfy.

What fell out. A word $x = b\,m\,c$ with outer bits b, c and middle m of width $L-2$ is a representative exactly when

$$(b,c) = (0,1), \qquad \text{or } (b,c) = (0,0) \text{ and } m \le Jm, \qquad \text{or } (b,c) = (1,1) \text{ and } m < Jm,$$

and the frame $(1,0)$ admits nothing (RUNS on all 131,064 words, $L = 3$ to 16). The grey frames 01 and 10, a light beside a gap, decide at once. The pure frames 00 and 11 pass the decision inward to the middle. A palindrome meets only pure frames all the way in, so it is decided by its outermost bit through the complement. Written as providers,

$$\text{reps}(L) = 0 \cdot W_{L-2} \cdot 1 \ \sqcup\ 0 \cdot \text{reps}(L-2) \cdot 0 \ \sqcup\ F\big(\text{reps}(L-2)\big), \qquad F(a) = \min\big(1a1,\ P(1a1)\big),$$

where W_{L-2} is the whole word layer, representatives and partners together. The three blocks are disjoint and exhaust the representatives (RUNS, $L = 3$ to 16). Every representative of width L therefore has exactly one provider in ring $L-2$, and the DI map is explicit: $0m1$ is provided by the word m wherever it sits; $0a0$ by the representative a; $1a1$ by the non-palindromic representative a through the reversal branch of F; and $0C(a)0$ by the palindromic representative a through the complement branch. The even rings descend from the 2-bit ring and the odd rings from the 1-bit ring, both through the same four 2-bit frames. "All derived from 2 bits" is this statement.

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5. Route 4. The folded square

The held state. The square of Section 1.

The trace. Place every word at (front half, reversed back half) and read the rules of Route 3 as geometry.

What fell out. Representatives are the cells above the diagonal plus the diagonal cells with $\varphi < 1/2$. The partner of a non-palindrome is the mirror cell. A palindrome (φ, φ) goes to the diagonal antipode $1 - \varphi - 2^{-n}$ (RUNS for every even width to 16). The frame of a word is its quadrant, because b is the top bit of f and c is the top bit of $J(g)$. Quadrant 01 lies wholly above the diagonal (yes), quadrant 10 wholly below (no), and quadrants 00 and 11 straddle it and each hold the whole square again at half scale (maybe). That is how yes and no merge into maybe: the undecided quadrants repeat inward, the undecided region shrinks by half at every step, and in the end only the diagonal is left. The diagonal is where a word equals its own reflection, so reflection cannot decide it and the complement does.

The centre of the diagonal is the largest palindromic representative $0\,1^{L-2}\,0$, the cell nearest $(1/2, 1/2)$ (RUNS). That is the cell where the previous session's pad law first breaks, so the pad-law edge is the fold of the diagonal.

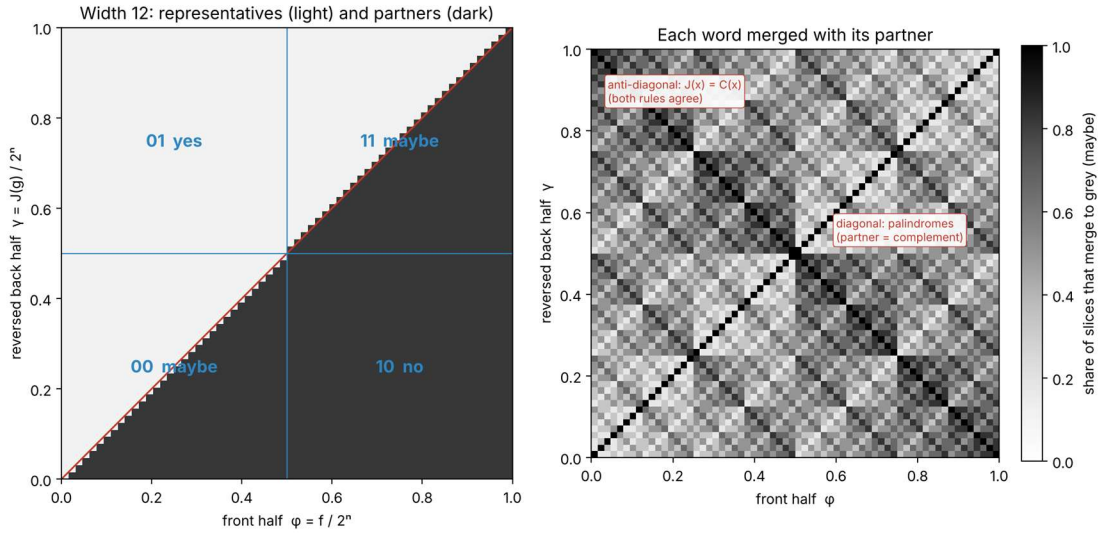


Figure 2. The folded square at width 12. Left: representatives (light) fill the triangle above the diagonal; partners (dark) are its mirror. The quadrant labels are the four frames. Right: the share of each word's slices that turn grey when it is merged with its partner. Grey is total on the diagonal (palindromes, partner by complement) and on the anti-diagonal (words whose reversal is their complement, where both rules give the same partner).

6. Route 5. The slope

The held state. The representative density is a straight line.

The trace. Average the frame decision over a free tail. Take a word whose value begins with a prefix q and leave the rest free. Pair h decides "representative" when its front bit is 0 and its mirror bit comes up 1, and

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every earlier pair deferred. The probability is $\sum_h [q_h = 0]\ 2^{-(h+1)}$ plus the deferral remainder, and the sum is $1 - \mathrm{mid}(q)$.

What fell out. The density of representatives at value t is $1 - t$. The finite form is exact. Write $N_L(q)$ for the number of representatives of width L whose value begins with the prefix q. For a prefix of length $p \geq 1$ with $2p \leq L$,

$$N_L(q) = 2^{L-p}\big(1 - \mathrm{mid}(q)\big) + 2^{\lceil (L-2p)/2 \rceil}\left([q_0 = 0] - \frac{1}{2}\right),$$

with no remainder, for all 3,030 prefix blocks with $L \leq 18$ (RUNS). Integrating the line gives the angle of a representative of value $t = a/2^L$:

$$u \approx t - \frac{t^2}{2}, \qquad \frac{1}{2} - u \approx \frac{(1-t)^2}{2}.$$

Two exact terms ride on the parabola: the tent below, and a back-half count that Route 7 writes out.

The seam is the apex of this parabola, the value where the density reaches zero. The largest deviation from the parabola over all representatives is below $2^{-L/2+1}$ at every width from 10 to 20 (RUNS); at width 20 it is 2.44×10^{-4}, just under 2^{-12}.

What the parabola leaves is a tent. Route 7 gives it exactly: the front-half count is $Q(f) = f\,2^n - f^2/2 + 1/2\,\min(f, 2^n - f)$, so the correction rises linearly from 0, peaks at $\varphi = 1/2$ and falls back to 0 (RUNS for every $n \leq 11$). The tent is half the palindromes, and its apex is the midpoint of the diagonal. The parabola is the fold of the square; the tent is the fold of the diagonal.

Constants fall out of the parabola directly. The frame 11 begins at $u(1/2) = 3/8$. The word at the quarter turn has value $1 - 1/\sqrt{2}$, and the residual gap times $2^{L/2}$ converges to $\left(\sqrt{2} - 1\right)/2 = 0.207107$, which is the tent at that value divided by the density there (RUNS at widths 40, 80, 160 and 320, in 120-digit arithmetic). The representatives' centre of mass is $\int t(1-t)\,dt / \int (1-t)\,dt = 1/3$, and the partners' is $2/3$; at width 20 they read 0.333091 and 0.666908 (RUNS). Inside a representative, bit h leans toward 0 and its mirror bit toward 1 by the same amount: $\Pr(x_h = 1) \to 1/2 - 2^{-(h+2)}$ and $\Pr(x_{L-1-h} = 1) \to 1/2 + 2^{-(h+2)}$. At width 20 the front bits read $0.24951,\ 0.375,\ 0.4375,\ 0.46875,\ 0.48438$ (RUNS). The representative tilts dark-to-light, front to back, which is the 01 frame written as a gradient. The "triangular top" near the seam, where blocks hold counts linear in their distance from the seam, is the same line seen at its zero.

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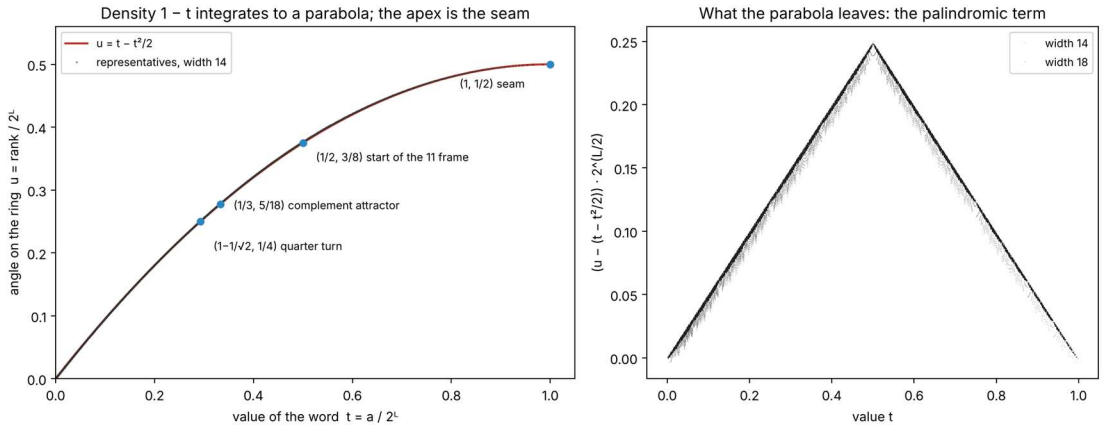


Figure 3. Left: every representative of width 14 at (value, angle) on the parabola $u = t - t^2/2$, with the four marked constants. Right: what the parabola leaves, scaled by $2^{L/2}$, is the tent $1/2 \min(t, 1-t)$ with its apex at the midpoint of the diagonal.

7. Route 6. Continuation and its two attractors

The held state. Continuation has two branches, and each has a fixed point in the square.

The trace. Strip frames from the outside until no provider is left. What remains is the root; the number of frames stripped is the depth.

What fell out. On the reversal branch $F(a) = 1a1$: the front becomes $1 \parallel f$ and the reversed back becomes $1 \parallel J(g)$. In the square that is the homothety of ratio $1/2$ toward the apex corner $(1,1)$ (RUNS). Its image is a quarter of the triangle, and one provider in four is continuation, so

$$\Pr(\text{depth} \geq t) = 4^{-t} \quad (1 \leq t \leq \lfloor (L-1)/2 \rfloor), \qquad \text{roots} = 3 \cdot 2^{L-3}, \qquad \mathbb{E}[\text{depth}] = \frac{1 - 4^{-\lfloor (L-1)/2 \rfloor}}{3},$$

exactly at every width to 16; at width 16 the mean is $5461/16384$ (RUNS). The mean tends to $1/3 = 0.0101\ldots_2$. In angle the reversal branch is the contraction $u \mapsto 3/8 + u/4$ to within $2^{-L/2}$, whose fixed point is the seam, and every reversal-branch word of depth d lies in the band $1/2 - 4^{-d}/2 \leq u < 1/2$ (RUNS for $L = 12,14,16$).

On the complement branch $F(a) = 0\,C(a)\,0$: a palindrome stays on the diagonal and moves by

$$\varphi \mapsto \frac{1-\varphi}{2},$$

exactly $f \mapsto (2^n - 1 - f)/2^{n+1}$ on the front half.

Its fixed point is $\varphi = 1/3$, the 2-cycle point of the doubling map. The complement ray of the root oo runs 00, 0110, 010010, 01011010, and so on, converging in value to $1/3$ and in angle to $u(1/3) = 1/3 - 1/18 = 5/18$. Followed to width 120, it reads 0.277777777778 and 0.333333333333 (RUNS). Its partner sits at angle $13/18$ with value $2/3$. The 01 ray converges to the seam: at width 120 its distance from angle $1/2$ is 2^{-120} (RUNS).

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Both branches have closed forms, so the codec never iterates. A non-palindromic root gives $F^k(\rho) = 1^k \rho\, 1^k$. A palindromic root gives $F^k(\rho) = a_k\, X\, a_k^R$, where a_k is the first k bits of $0101\ldots$ and X is ρ or $C(\rho)$ according to the parity of k. The record (ρ, k, ε), with ε the partner side, round-trips every one of the 8,190 nonempty words of width at most 12 (RUNS), and the partner acts on it by flipping ε alone.

The constant $1/3 = \sum_{k\geq 1} 4^{-k} = 0.0101\ldots_2$ has now landed three times: as the mean depth, as the complement fixed point, and inside the representatives' centre of mass, $1/2 - 1/2\sum_{k\geq 1} 4^{-k}$. These are one path read at three locations. Every step inward through a frame enters a quadrant, a quarter of the square, and a quarter per step sums to $1/3$: in the count of providers, on the diagonal (two complement steps are $\varphi \mapsto (1+\varphi)/4$), and in the bit bias that sets the centre of mass.

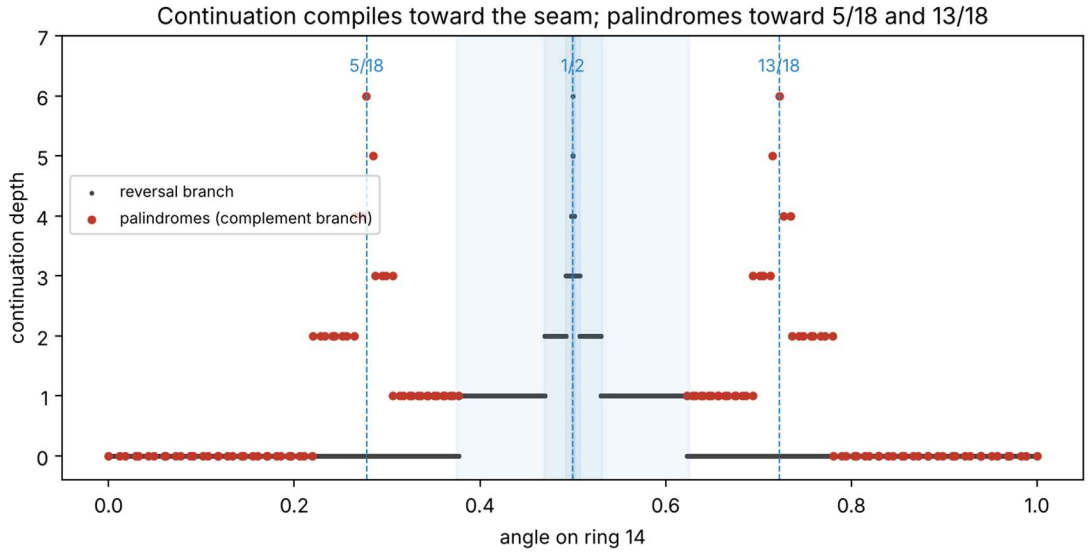


Figure 4. Continuation depth of every word of ring 14 against its angle. The reversal branch compiles toward the seam in nested bands of half-width $4^{-t}/2$; the palindromes on the complement branch close in on $5/18$ and $13/18$ from alternating sides.

8. Route 7. The address

The held state. The ring is addressed by a quadratic in the front half and a count in the back half.

The trace. Count the representatives column by column across the square, then inside the column.

What fell out. For even $L = 2n$ and a representative $x = f \parallel g$,

$$\operatorname{rank}(x) = f(2^n - 1) - \frac{f(f-1)}{2} + \min(f, 2^{n-1}) + D_f(g) + [f < 2^{n-1}]\,[J(f) < g],$$

where $D_f(g)$ counts the back halves $g' < g$ whose reversal exceeds the front, $J(g') > f$.

The first three terms are $Q(f)$, the number of representatives in the columns to the left: the triangle's area plus the tent. The fourth, $D_f(g)$, is a bit-reversal dominance count with a one-step recursion per bit. For odd $L = 2n + 1$ with middle bit c,

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$$\mathrm{rank}(x) = 2Q(f) + c\left(2^n - \theta_f\right) + D_f(g) + [f < 2^{n-1}]\,[J(f) < g], \qquad \theta_f = f + [f \geq 2^{n-1}].$$

Both forms are exact for all 65,534 representatives with $L \leq 16$ and agree with the frame-peeling rank on random words to width 512 (RUNS). Unranking inverts the quadratic with one integer square root and an $O(1)$ correction, then selects the back half bit by bit, with an $O(1)$ count per bit: the number of completions of a top prefix p with $J(g) \geq \theta$ is $2^j - \min\left(\max\left(\lceil\left(\theta - J(p)\right)/2^{n-j}\rceil, 0\right),\ 2^j\right)$. The engine from the previous session searched a digit automaton by bisection, which takes $O(L^2)$ steps. This address takes $O(L)$. It is exact at every position for $L \leq 14$, round-trips random positions to width 1024, and agrees with the reference at every tested position to width 256 (RUNS). At width 512 it returns a word in milliseconds where the reference takes seconds, more than three orders of magnitude faster. A 4096-bit ring word comes back, with its location, in about 12 to 20 ms.

The radial reader inherits the square root. At angle $u < 1/2$ and ring L, the reader lands in the word of rank $\lfloor 2^L u \rfloor$, and that word's front half is the first $L/2$ bits of

$$\tau(u) = 1 - \sqrt{1 - 2u}.$$

Every reading that falls in the front half of its word is a digit of that square root. Over 3,892 sampled front-half readings with L from 24 to 160, 3,891 agree (RUNS, sealed threshold 95%). The back half carries the mirror's answer instead. The reversed back half of a representative is a uniform draw above its front half, $\varphi \leq J(g)/2^n < 1$, so back-half readings agree with τ only at chance away from the pole and the seam. The ring is read by a continuum in its front half and by a count in its back half, half a word apart.

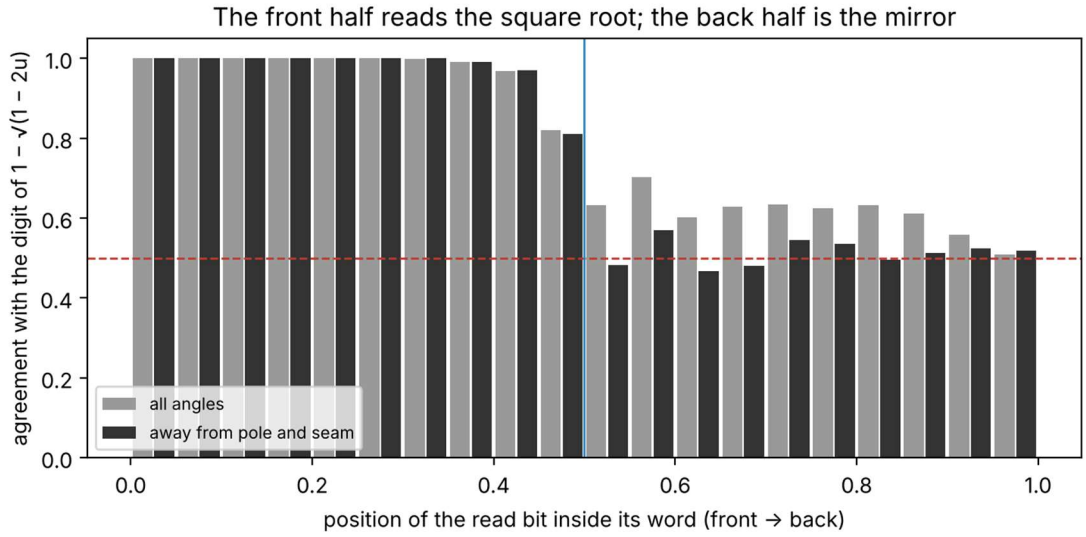


Figure 5. Agreement between the radial reading and the digit of $1 - \sqrt{1 - 2u}$ at the same position, by where the reading falls inside its word. The front half reads the square root; the back half does not.

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9. Route 8. The merge

The held state. Lay each word on its partner. Where both are 1 the slice stays light, where both are 0 it stays a gap, and where they disagree it turns grey. The grey pattern of a pair is its residual $x \oplus P(x)$, its maybe-mask, and the mask is always a palindrome.

The trace. In the square, a non-palindrome's residual is $d \parallel J(d)$ with $d = f \oplus J(g)$. Grey is the XOR distance between a word's two coordinates, written out and mirrored.

What fell out. Summed over every word of width L,

$$\sum_x |x \oplus P(x)| = \left\lfloor \frac{L}{2} \right\rfloor \cdot 2^L + L \cdot 2^{\lceil L/2 \rceil},$$

and the slices where the pair agrees split exactly evenly between light and gap (RUNS for $L \le 16$). The grey share is therefore

$$g_L = \frac{\lfloor L/2 \rfloor \cdot 2^L + L \cdot 2^{\lceil L/2 \rceil}}{L \cdot 2^L} = 1,\ 1,\ 5/6,\ 3/4,\ 13/20,\ 5/8,\ 31/56,\ 9/16,\ 73/144,\ 17/32,\ \ldots \to \frac{1}{2}.$$

At two bits the merge is total: 00 meets 11 and 10 meets 01, and every slice is grey. A bunch of yes and no merge into maybe completely at two bits, and into exactly half maybe in the limit. The two tilts of Route 5 cancel in the merge: a representative leans dark-to-light and its partner leans the other way, so the merged pair has no direction left. Grey is total for a word exactly when it is a palindrome, partnered by complement, or an anti-palindrome with $J(x) = C(x)$, where the two rules give the same partner. There are $2^{\lceil L/2 \rceil} + [L \text{ even}]\ 2^{L/2}$ such words, on the diagonal and the anti-diagonal of the square (RUNS, $L \le 16$).

The light is balanced at every width. Every ring holds exactly $L \cdot 2^{L-1}$ ones in its $L \cdot 2^L$ slices (RUNS, $L \le 14$). Its partner half carries exactly $2^{\lceil L/2 \rceil}$ more ones than its representative half, which is the number of palindromes of width L, for every $L \ge 2$ (RUNS). At $L = 1$ the halves differ by 1, not 2: the sealed line said "each ring", and it failed at the one ring whose only palindrome is its own middle bit (DEAD END, routed to the 1-bit ring).

The previous session's results land in the same mirror. The residual alphabet holds $2^m - 1$ distinct masks at even width $2m$ and 2^m at odd width $2m + 1$, because the masks are the nonzero XOR distances from the diagonal (STANDS from the prior session; re-run here). The smallest width whose merge can carry a full byte is therefore 17, and all 256 bytes survive the merge there (re-run here).

10. Route 9. The seam

The held state. The seam is the apex of the parabola and the corner of the square, read from both sides.

The trace. Approach the half turn from each side and read one bit per ring.

What fell out. Just right of the seam the radial reading is $1,0,0,1,1,1,\ldots$; just left it is $0,0,0,1,1,1,\ldots$ (RUNS to ring 24). The two sides differ only at ring 1, which is yes on one side and no on the other, and agree on every deeper ring. The ordinary angle $1/2$ cannot hold the difference; the two-sided address $1/2^{\pm}$ of the previous session's inverse limit holds it. At even width $2a + 2$ the word just left of the seam is $1^{a+1}\,0\,1^a$ and the word

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just right of it is $1^a\,0\,1^{a+1}$. The largest representative $1^a\,0\,1^{a+1}$ sits just right of the seam because of the swap, and its reversal sits just left. At the apex, where the density has fallen to zero, the last pair exchanges places. The 01 ray $1^k\,0\,1^{k+1}$ with residual $0^k\,11\,0^k$ is this seam word at every even width, and the pad law $W_{L+2}(j) = 1\,W_L(j)\,1$ holds for $0 \le j < K_L = 1 + \left(2^{L-2} - 2^{\lceil (L-2)/2 \rceil}\right)/2$ and first fails at K_L (STANDS from the prior session; re-run here for $L = 2$ to 11).

11. Route 10. The radial language

The held state. A radial reading at depth D is a word of D bits, one per ring. The question carried over from the previous session is how many distinct words the rings let a ray read.

The trace. Merge every slice boundary of rings 1 to D on one integer grid, read one bit per ring in every cell, and count distinct profiles R_D together with the angular entropy.

What fell out. The rerun reproduces the earlier table exactly, R_{10} to $R_{16} = 924,\ 1827,\ 3606,\ 7122,\ 14087,\ 27819,\ 55122$, with $H_{\text{angle}}(16) = 14.951476$ bits (RUNS). The same function carries it to $R_{17} = 109200$, $R_{18} = 216428$, $R_{19} = 429416$ and $R_{20} = 852130$. The deficit added per ring fell from 0.0183 at $D = 15$ to $0.0134,\ 0.0137,\ 0.0131,\ 0.0115,\ 0.0113$ at $D = 16$ to 20. The trend points toward a sublinear deficit and a growth rate of 2. Route 7 supplies the handle the next link needs: in the front half of every word a ray reads digits of $\tau(u)$ at positions set by the ray's own tail. The asymptotic law stays OPEN.

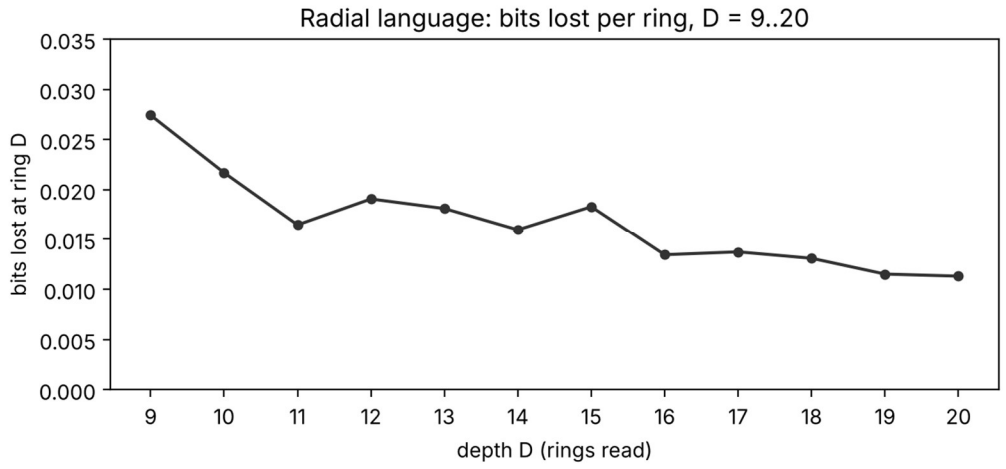


Figure 6. Bits lost per ring by the radial language, depths 9 to 20.

12. The map and the constants

Each route handed the next one the provider it needed: the stack gave the frame, the frames and the antipode gave the square, and the square gave the slope, the continuation and the merge. The slope gave the address and the apex, and the address and the apex gave the radial reading. Drawn as a dependency chain, the routes are one program.

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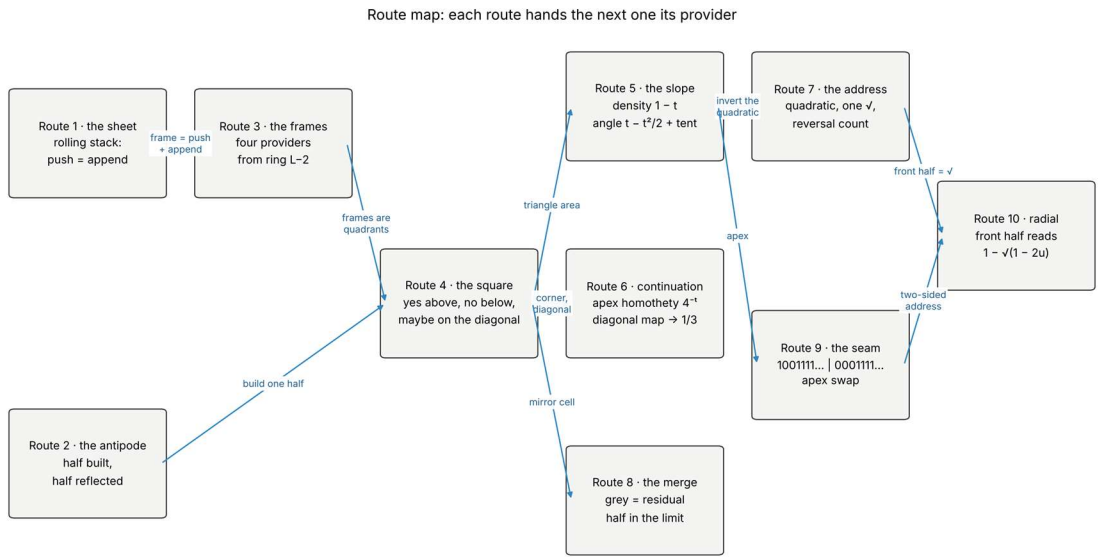


Figure 7. Route map.

Constant	Value	Where it falls out
Representative density at value t	$1 - t$	Route 5, block law
Angle of a representative of value t	$t - t^2/2$ plus tent	Route 5, Route 7
Start of the 11 frame	$3/8$	parabola at $t = 1/2$
Seam	$1/2$	apex, density zero
Quarter-turn word	$1 - 1/\sqrt{2}$; gap $\times\ 2^{L/2} \to (\sqrt{2} - 1)/2$	parabola and tent
Complement attractor	value $1/3$, angle $5/18$; partner $2/3$, $13/18$	diagonal map $\varphi \mapsto (1 - \varphi)/2$
Mean continuation depth	$1/3$	one provider in four
Root share, continuation share	$3/4$, $1/4$	four providers
Centre of mass, reps and partners	$1/3$, $2/3$	area under the line
Bit bias h places from an end	$\mp 2^{-(h+2)}$, front and mirror	frame decision
Light share of every ring	exactly $1/2$	every word once
Half imbalance	$2^{\lceil L/2 \rceil}$ ones	palindromes
Grey share of the merge	$g_L \to 1/2$; total at $L = 2$	XOR distance
Totally grey words	$2^{\lceil L/2 \rceil} + [L \text{ even}]\ 2^{L/2}$	diagonal and anti-diagonal
Twisted slices	$2^{\lceil L/2 \rceil}/2^L$	antipode
Pad-law edge	$0\ 1^{L-2}\ 0$	centre of the diagonal
Seam profiles	$1001111\ldots$ and $0001111\ldots$	two-sided address
Sheet chain	0 0 1 3 6 13 27 54 108 216	rolling stack

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13. PULL: the constraint chain read off the ring

The constraint chain Dean Kulik is building runs from "all things must change" toward a third law, and the ring supplies exact counterparts for most of its links. The mathematics in this section all RUNS. Each identification with the chain is a PULL.

No word holds its place without a partner to move to, because P has no fixed point; this is Law 2 in its smallest form. The new location inherits: ring L is provided by ring $L-2$ through four frames, and the pad law carries the seam word outward unchanged. The price of a transformation is a gradient and not a single value: the representative density is the line $1-t$. There is direction: inside every representative, light leans from front to back by $2^{-(h+2)}$ at the bits h places from each end. All change is equal: the density loses the same amount for every equal step in value, so the second derivative of angle against value is the constant -1. Collision: at the apex the last pair exchanges places.

Read off this ring, Law 3 is the following candidate. **Equal change makes location quadratic in value: the distance to the apex in location is half the square of the distance to the apex in value,** $1/2-u=1/2\,(1-t)^2$**. The apex is where the price reaches zero, and there the last pair exchanges places.** The fold of the diagonal adds one term, the tent, which is half the palindromes. Which factor of the constant -1 is mass and which is acceleration is not written by any route here; that link is OPEN.

The dual wave has a counterpart too: the front half of every word is read by a continuum (the square root) and the back half by a count (the mirror). The two are offset by half a word and never read concurrently. That is a PULL to the compiler and reader waves of *The Inverted Mold* and *The Reading of Potential*.

14. Ledger

Status	Claim
RUNS	Sheet rebuilt cell by cell (896 cells); rolling stack, two providers agree on every tape; $G_0(W)=B_0(W+3)$; nibble law; half turn is the nibble swap; 2-bit reflection at every width
RUNS	Past two bits the reversed tape is never a rotation; the half turn at circumference swaps halves on ring 3
RUNS	Antipode $R[N-1-i]=P(R[i])$; tape twist confined to palindromic words; sheet reading is the tape with non-palindromic words complemented; counting only at two bits
RUNS	Frame law and four providers, $L=3$ to 16
RUNS	Folded square: triangle, mirror, diagonal antipode, frames as quadrants, pad-law edge at the diagonal's centre
RUNS	Block law (3,030 blocks); parabola within $2^{-L/2+1}$; tent identity; quarter-turn value $1-1/\sqrt{2}$; bit bias; centres of mass $1/3$ and $2/3$
RUNS	Depth law 4^{-t}, roots $3\cdot 2^{L-3}$, mean depth; closed-form codec; complement attractor $1/3$ at $5/18$; reversal ray to the seam
RUNS	Exact split rank, even and odd; square-root unrank; $O(L)$ address to width 4096; radial reader reads $1-\sqrt{1-2u}$ (3,891 of 3,892)
RUNS	Merge accounting; light equals gap; total grey on diagonal and anti-diagonal; every ring half light; halves differ by $2^{\lfloor L/2 \rfloor}$ for $L\geq 2$
RUNS	Seam profiles $1001111\ldots$ and $0001111\ldots$; radial table rerun exactly and extended to $D=20$

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Status	Claim
STANDS	Prior session, re-run here: pad-law range K_L; residual alphabet $2^m - 1$ and 2^m; byte codec at width 17; 01 ray; radial table to $D = 16$
STANDS	Prior session, not re-run: the inverse-limit completion of the angular address with sided addresses $u^{\pm}$ (standard facts about inverse limits of finite partitions)
STANDS	Prior audit: the partner rule is one fixed-point-free involution among several; Gray code, reciprocal CRC and byte-reverse instructions share reflection but do not implement the conditional partner; a residual alone does not carry its word's address
DEAD END	S7 as sealed ("each ring") at $L = 1$: the halves differ by 1. Owner: the 1-bit ring, whose only palindrome is its own middle bit
DEAD END	Previous session: "the rings are counting". True of the 2-bit sheet only. Owner: the sheet reading at $L = 2$ (Route 2); the two-hole law and $0.\overline{C_n} = 1/(B-1)^2 - B/\big((B-1)(B^B-1)\big)$ belong to the circular counter, not to the partner ring
DEAD END	Previous session's Law 3 candidate, "equal change at width n forces exactly two forbidden readings at $n+1$". Uniform occupancy does not force two holes: 00010111 has uniform 2-bit occupancy and reads all eight 3-bit words. Owner: the counter's carry. Withdrawn; Section 13 replaces it
OPEN	Growth rate of the radial language: the per-ring deficit is falling to $D = 20$, and the limit is not written
OPEN	The uniform witness family for the radial profile $1\,0\,0\,1^{D-5}\,0\,0$; the core of the seam word at offset 2^q is the missing link, now reachable through the split rank
OPEN	Which factor of the constant second derivative is mass and which is acceleration
PULL	The constraint chain of Section 13; the dual-wave reading of front and back halves

15. Reproducibility

The notebook nested_ring_routes.ipynb runs top to bottom in about 30 seconds on two cores, with two_bit_circle.xlsx beside it for Route 1, and prints the ledger above. The radial counts at depths 17 to 20 use the notebook's radial_counts function run with Dmax=20 (about two minutes and 2 GB). The predictions were sealed before their checks ran, and the SHA-256 hashes of the seal files are:

Seal	SHA-256
seal_routes.txt	18cbef3f425c5a51c727a59a4dacb9cdc43b6c9bf24771ec5eeace1c0ae7163f
seal_gradient.txt	721fafeb65c80801209d7ef47ba41fde69ad679d7722fe3b9e02ee0dab9b317d
seal_address.txt	3cc4adb2e2c4ad1a75935521c390c9413cfa9abdd6933a580d4f6903341467ec
seal_square.txt	4b23106de75067276c84b9ddd5a8cbee3a15fd2a61dfbc0fbe27f62f8ec8a80f
seal_merge.txt	325c8204b335d435194bea9ec22cb18294932e8dbc127124a2e9c6a1eecf78f6

The companion workbook nested_ring_beyond_2bit.xlsx carries the window sheet for rings 3 and 4 in the original layout, and a DI-chain sheet that lists every word of rings 1 to 10 with its partner, frame, provider, residual, depth and its angle against the parabola.

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Prior AI sessions in the NEXUS peer network (October 2026): the unbounded ring engine (unbounded_ring_engine.py), the pad-law range, the residual alphabet, the root–depth–side codec and the radial-language counts, re-run in this paper.

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